

Theo SOULAT

# **Mise en place d'un Cluster Web Haute Disponibilité**

HAProxy + Heartbeat

BTS SIO — Option SISR

2025-2026

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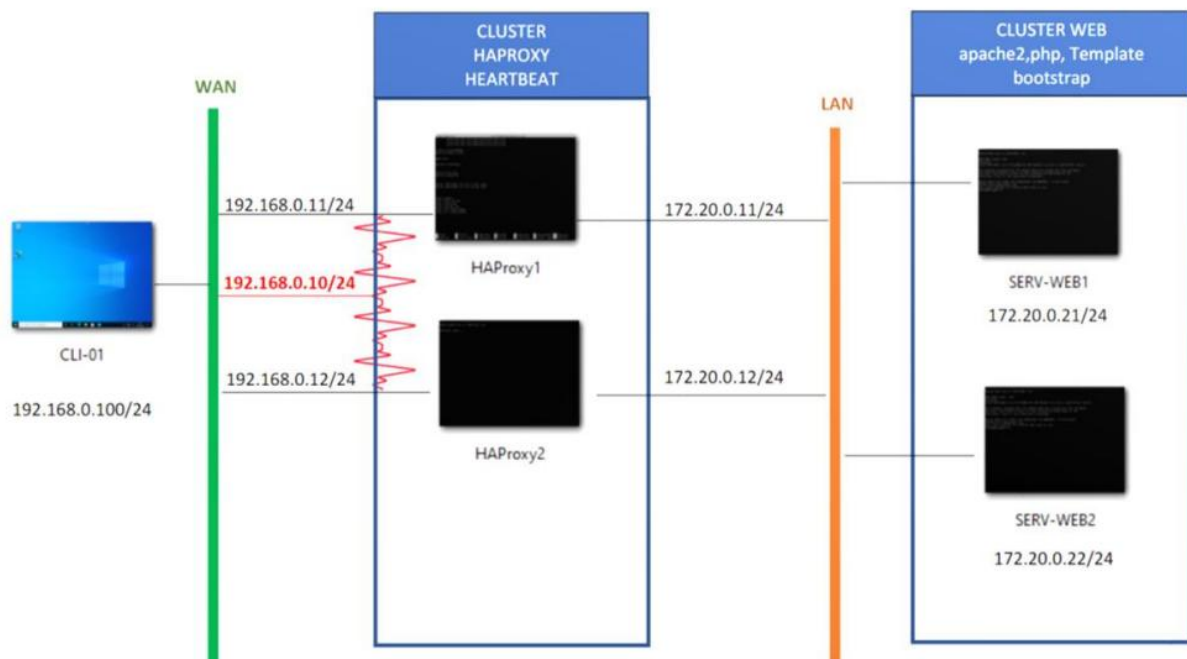
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# Introduction

Dans le monde professionnel actuel, les services web doivent être disponibles en permanence. Une interruption de service, même de courte durée, peut entraîner des pertes financières importantes et nuire à l'image d'une entreprise. Il est donc essentiel de mettre en place des solutions garantissant la continuité de service : c'est ce que l'on appelle la haute disponibilité.

## Architecture de l'infrastructure

Schéma :



## Table des machines virtuelles :

Le tableau suivant récapitule les caractéristiques de chaque VM à créer :

| VM        | Rôle                      | IP WAN (eth0)    | IP LAN (eth1)  | RAM / vCPU |
|-----------|---------------------------|------------------|----------------|------------|
| CLI-01    | Client de test            | 192.168.0.100/24 | —              | 1 Go / 1   |
| HAProxy1  | Load balancer primaire    | 192.168.0.11/24  | 172.20.0.11/24 | 1 Go / 1   |
| HAProxy2  | Load balancer secondaire  | 192.168.0.12/24  | 172.20.0.12/24 | 1 Go / 1   |
| SERV-WEB1 | Serveur web (Apache2/PHP) | —                | 172.20.0.21/24 | 1 Go / 1   |
| SERV-WEB2 | Serveur web (Apache2/PHP) | —                | 172.20.0.22/24 | 1 Go / 1   |

## Plan d'adressage réseaux

| Réseau     | Sous-réseau    | Type VMnet        | Machines connectées                      |
|------------|----------------|-------------------|------------------------------------------|
| <b>WAN</b> | 192.168.0.0/24 | VMnet (Host-only) | CLI-01, HAProxy1, HAProxy2               |
| <b>LAN</b> | 172.20.0.0/24  | VMnet (Host-only) | HAProxy1, HAProxy2, SERV-WEB1, SERV-WEB2 |

# Démarche de réalisation

## Étape 1 – Préparation VMware

Configuration IP de chaque VM :

### HAProxy1

```
root@debian:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:9e:25:4e brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altnam enx000c299e254e
    inet 192.168.0.11/24 brd 192.168.0.255 scope global ens33
        valid_lft forever preferred_lft forever
    inet 192.168.0.129/24 brd 192.168.0.255 scope global secondary dynamic noprefixroute ens33
        valid_lft 1335sec preferred_lft 1110sec
    inet6 fe80::183a:6a80:1e4a:4302/64 scope link
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:9e:25:58 brd ff:ff:ff:ff:ff:ff
    altnam enp2s5
    altnam enx000c299e2558
    inet 172.20.0.11/16 brd 172.20.255.255 scope global ens37
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe9e:2558/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
root@debian:~# _
```

### HAProxy2

```
root@debian:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:74:79:dc brd ff:ff:ff:ff:ff:ff
    altnam enp2s1
    altnam enx000c297479dc
    inet 192.168.0.12/24 brd 192.168.0.255 scope global ens33
        valid_lft forever preferred_lft forever
    inet 192.168.0.128/24 brd 192.168.0.255 scope global secondary dynamic noprefixroute ens33
        valid_lft 1640sec preferred_lft 1415sec
    inet6 fe80::2c19:ba81:358a:fa02/64 scope link
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:74:79:e6 brd ff:ff:ff:ff:ff:ff
    altnam enp2s5
    altnam enx000c297479e6
    inet 172.20.0.12/16 brd 172.20.255.255 scope global ens37
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe74:79e6/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
root@debian:~# _
```

## SERV-WEB1

```
root@debian:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:76:df:29 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c2976df29
    inet 192.168.254.128/24 brd 192.168.254.255 scope global dynamic noprefixroute ens33
        valid_lft 1017sec preferred_lft 792sec
    inet6 fe80::18b4:ea4d:a6fa:71fa/64 scope link
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:76:df:33 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    altname enx000c2976df33
    inet 172.20.0.21/16 brd 172.20.255.255 scope global ens37
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe76:df33/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
root@debian:~# _
```

## SERV-WEB2

```
root@debian:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c5:aa:37 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29c5aa37
    inet 192.168.254.129/24 brd 192.168.254.255 scope global dynamic noprefixroute ens33
        valid_lft 1085sec preferred_lft 860sec
    inet6 fe80::e96f:acaf:81e3:4df9/64 scope link
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c5:aa:41 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    altname enx000c29c5aa41
    inet 172.20.0.22/16 brd 172.20.255.255 scope global ens37
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fec5:aa41/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
root@debian:~#
```

## CLI-01

```
📄 Invite de commandes
Microsoft Windows [version 10.0.19045.2965]
(c) Microsoft Corporation. Tous droits réservés.

C:\Users\pc47>ipconfig

Configuration IP de Windows

Carte Ethernet Ethernet0 :

    Suffixe DNS propre à la connexion. . . . :
    Adresse IPv6 de liaison locale. . . . . : fe80::116b:2ea8:39c6:c98c%8
    Adresse IPv4. . . . . : 192.168.0.100
    Masque de sous-réseau. . . . . : 255.255.255.0
    Passerelle par défaut. . . . . :

C:\Users\pc47>
```

## ip route sur HAProxy1

```
root@debian:~# ip route
172.20.0.0/16 dev ens37 proto kernel scope link src 172.20.0.11
192.168.0.0/24 dev ens33 proto dhcp scope link src 192.168.0.129 metric 1002
root@debian:~# _
```

## Ping de HAProxy1 vers SERV-WEB1

```
root@debian:~# ping 172.20.0.21
PING 172.20.0.21 (172.20.0.21) 56(84) bytes of data.
64 bytes from 172.20.0.21: icmp_seq=1 ttl=64 time=0.577 ms
64 bytes from 172.20.0.21: icmp_seq=2 ttl=64 time=0.401 ms
64 bytes from 172.20.0.21: icmp_seq=3 ttl=64 time=0.477 ms
64 bytes from 172.20.0.21: icmp_seq=4 ttl=64 time=0.303 ms
64 bytes from 172.20.0.21: icmp_seq=5 ttl=64 time=0.354 ms
64 bytes from 172.20.0.21: icmp_seq=6 ttl=64 time=0.411 ms
64 bytes from 172.20.0.21: icmp_seq=7 ttl=64 time=0.347 ms
^C
--- 172.20.0.21 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6150ms
rtt min/avg/max/mdev = 0.303/0.410/0.577/0.085 ms
root@debian:~#
```

## Ping de CLI-01 vers HAProxy2

```
C:\Users\pc47>ping 192.168.0.12

Envoi d'une requête 'Ping' 192.168.0.12 avec 32 octets de données :
Réponse de 192.168.0.12 : octets=32 temps<1ms TTL=64
Réponse de 192.168.0.12 : octets=32 temps<1ms TTL=64
Réponse de 192.168.0.12 : octets=32 temps<1ms TTL=64
Réponse de 192.168.0.12 : octets=32 temps<1ms TTL=64

Statistiques Ping pour 192.168.0.12:
    Paquets : envoyés = 4, reçus = 4, perdus = 0 (perte 0%),
Durée approximative des boucles en millisecondes :
    Minimum = 0ms, Maximum = 0ms, Moyenne = 0ms

C:\Users\pc47>
```

## Étape 2 – Serveurs Web

On peut voir que apache2 fonctionne car il est actif.

```
root@debian:~# systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-04-20 22:57:30 CEST; 2h 5min ago
  Invocation: 4ef9b46098574b32bf0b5380a37e8410
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 928 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Process: 1533 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=0/SUCCESS)
    Main PID: 953 (apache2)
       Tasks: 6 (limit: 2255)
      Memory: 24.7M (peak: 32.5M)
         CPU: 1.139s
    CGroup: /system.slice/apache2.service
            └─ 953 /usr/sbin/apache2 -k start
               1542 /usr/sbin/apache2 -k start
               1543 /usr/sbin/apache2 -k start
               1544 /usr/sbin/apache2 -k start
               1545 /usr/sbin/apache2 -k start
               1546 /usr/sbin/apache2 -k start

avril 20 22:57:30 debian systemd[1]: Starting apache2.service - The Apache HTTP Server...
avril 20 22:57:30 debian apachectl[947]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive dynamically to determine the server's name. For more information, see the http://httpd.apache.org/docs/2.4/using\_server\_name.html file.
avril 20 22:57:30 debian systemd[1]: Started apache2.service - The Apache HTTP Server.
avril 21 00:49:24 debian systemd[1]: Reloading apache2.service - The Apache HTTP Server...
avril 21 00:49:24 debian apachectl[1536]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName' directive dynamically to determine the server's name. For more information, see the http://httpd.apache.org/docs/2.4/using\_server\_name.html file.
avril 21 00:49:24 debian systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
lines 1-25/25 (END)
```

Voici le contenu de /var/www/html/index.php

```
<!DOCTYPE html>
<html>
<head>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
  <title>Test HA - BTS SIO</title>
</head>
<body class="container text-center mt-5">
  <div class="card shadow">
    <div class="card-body">
      <h1 class="display-4 text-primary">Réponse du serveur :</h1>
      <h2 class="badge bg-success fs-2">SERV-WEB1 (172.20.0.21)</h2>
      <p class="lead mt-3">Infrastructure Haute Disponibilité HAProxy + Heartbeat</p>
    </div>
  </div>
</body>
</html>
```

curl <http://172.20.0.21> pour prouver que le serv-web1 marche

```
root@debian:~# curl http://172.20.0.21
srweb1
```

curl <http://172.20.0.22> pour prouver que le serv-web2 marche

```
root@debian:~# curl http://172.20.0.22
srweb2
```



## Étape 3 – Configuration HAProxy

**Frontend** : point d'entrée des requêtes clients, HAProxy écoute sur une IP et un port et redirige le trafic.

**Backend** : ensemble des serveurs web qui traitent les requêtes (SERV-WEB1 et SERVWEB2).

**Round-robin** : méthode qui répartit les requêtes de manière alternée et équilibrée entre les serveurs.

**Health checks** : vérifications automatiques de l'état des serveurs pour exclure ceux en panne.

**Page stats** : interface web permettant de visualiser l'état du frontend, des backends et du load balancing.

Voici le contenu complet de haproxy.cfg

```
GNU nano 8.4 /etc/haproxy/haproxy.cfg
global
    log /dev/log      local0
    log /dev/log      local1 notice
    chroot /var/lib/haproxy
    stats socket /run/haproxy/admin.sock mode 660 level admin
    stats timeout 30s
    user haproxy
    group haproxy
    daemon

    # Default SSL material locations
    ca-base /etc/ssl/certs
    crt-base /etc/ssl/private

    # See: https://ssl-config.mozilla.org/#server=haproxy&server-version=2.0.38&config=intermediate
    ssl-default-bind-ciphers ECDHE-ECDSA-AES128-GCM-SHA256:ECDHE-RSA-AES128-GCM-SHA256:ECDHE-ECDSA-AES256-GCM-SHA384:ECDHE-RSA-AES256-GCM-SHA384:ECDHE-ECDSA-
    ssl-default-bind-ciphersuites TLS_AES_128_GCM_SHA256:TLS_AES_256_GCM_SHA384:TLS_CHACHA20_POLY1305_SHA256
    ssl-default-bind-options ssl-min-ver TLSv1.2 no-tls-tickets

defaults
    log          global
    mode         http
    option       httplog
    option       dontlognull
    timeout connect 5000
    timeout client  50000
    timeout server  50000
    errorfile 400 /etc/haproxy/errors/400.http
    errorfile 403 /etc/haproxy/errors/403.http
    errorfile 408 /etc/haproxy/errors/408.http
    errorfile 500 /etc/haproxy/errors/500.http
    errorfile 502 /etc/haproxy/errors/502.http
    errorfile 503 /etc/haproxy/errors/503.http
    errorfile 504 /etc/haproxy/errors/504.http

listen clusterWeb
    bind *:80
    # mode d'ecoute
    mode http
    # mode du balancement (round robin (50%-50%))
    balance roundrobin
    # Option
    option httpclose
    option forwardfor
    # Liste des serveurs impliqués par le balancement
    server SRV-WEB1 172.20.0.21:80 check

server SRV-WEB2 172.20.0.22:80 check
# Pour les statistiques
stats enable
stats hide-version
stats refresh 30s
stats show-node
stats auth admin:password
stats uri /statistique
```

## Et voici systemctl status haproxy

```
root@debian:~# systemctl status haproxy
• haproxy.service - HAProxy Load Balancer
  Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; preset: enabled)
  Active: active (running) since Mon 2026-04-20 23:24:21 CEST; 1h 51min ago
  Invocation: 877e74deaff347ba83956ba8f3e68a37
  Docs: man:haproxy(1)
        file:/usr/share/doc/haproxy/configuration.txt.gz
  Main PID: 1245 (haproxy)
  Status: "Ready."
  Tasks: 3 (limit: 2255)
  Memory: 78.6M (peak: 79.6M)
  CPU: 3.861s
  CGroup: /system.slice/haproxy.service
          └─1245 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock
            1258 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock

avril 20 23:24:21 debian systemd[1]: Starting haproxy.service - HAProxy Load Balancer...
avril 20 23:24:21 debian haproxy[1245]: [NOTICE] (1245) : New worker (1258) forked
avril 20 23:24:21 debian systemd[1]: Started haproxy.service - HAProxy Load Balancer.
avril 20 23:24:21 debian haproxy[1245]: [NOTICE] (1245) : Loading success.
```

## La page de stats de HAProxy1

Statistics Report for HAProxy on x +

Non sécurisé | 192.168.0.11/statistique

### HAProxy

#### Statistics Report for pid 1338 on debian

> General process information

pid = 1338 (process #1, nbproc = 1, nbthread = 2)  
uptime = 00:00:03.000s, warnings = 0  
system limits: memmax = unlimited, ulimit-n = 524287  
maxsock = 524287, maxconn = 262117, reached = 0, maxpipes = 0  
current conn = 1, current pipes = 0, conn rate = 1/sec, bit rate = 0.000 kbps  
Running tasks: 0/18 (0 nice), idle = 100 %

active UP, active UP, going down, active DOWN, going up, active or backup DOWN, active or backup DOWN for maintenance (MAINT), active or backup SOFT STOPPED for maintenance, backup UP, backup UP, going down, backup DOWN, going up, not checked

Note: "NOLB/DRAIN" = UP with load-balancing disabled.

Display option: Scope: Hide DOWN servers, Disable refresh, Refresh now, CSV export, JSON export (schema)

External resources: Privacy site, Updates (v3.0), Online manual

| clusterWeb | Queue |     |       | Session rate |     |       | Sessions |     |       | Bytes  |       | Denied |    | Errors |     | Warnings |     | Status | LastChk | Server |      |      |         |             |     |     |     |     |          |        |   |
|------------|-------|-----|-------|--------------|-----|-------|----------|-----|-------|--------|-------|--------|----|--------|-----|----------|-----|--------|---------|--------|------|------|---------|-------------|-----|-----|-----|-----|----------|--------|---|
|            | Cur   | Max | Limit | Cur          | Max | Limit | Cur      | Max | Limit | Total  | LbTot | Last   | In | Out    | Req | Resp     | Req |        |         | Conn   | Resp | Retr | Redis   | Wght        | Act | Bck | Chk | Dwn | Downtime | Thrtle |   |
| Frontend   |       |     |       |              |     |       |          |     |       |        |       |        |    |        |     |          |     |        |         |        |      |      |         |             |     |     |     |     |          |        |   |
| SRV-WEB1   | 0     | 0   | -     | 0            | 0   | -     | 0        | 0   | -     | 0      | 0     | ?      | 0  | 0      | 0   | 0        | 0   | 0      | 0       | 0      | 0    | 0    | 3m5s UP | L4OK in 1ms | 1/1 | Y   | -   | 0   | 0        | 0s     | - |
| SRV-WEB2   | 0     | 0   | -     | 0            | 0   | -     | 0        | 0   | -     | 0      | 0     | ?      | 0  | 0      | 0   | 0        | 0   | 0      | 0       | 0      | 0    | 0    | 3m5s UP | L4OK in 0ms | 1/1 | Y   | -   | 0   | 0        | 0s     | - |
| Backend    | 0     | 0   | -     | 0            | 0   | -     | 0        | 0   | -     | 26 212 | 0     | 0      | 0s | 0      | 0   | 0        | 0   | 0      | 0       | 0      | 0    | 0    | 3m5s UP |             | 2/2 | 2   | 0   | 0   | 0        | 0s     |   |

## Plusieurs tests depuis le CLI-01 de curl <http://192.168.0.11> IP

```
C:\Users\pc47>curl http://192.168.0.12
srvweb2

C:\Users\pc47>curl http://192.168.0.12
srvweb1

C:\Users\pc47>curl http://192.168.0.11
srvweb2

C:\Users\pc47>curl http://192.168.0.11
srvweb1
```

## Étape 4 – Configuration Heartbeat

**Principe actif/passif** : un seul nœud HAProxy est actif à la fois et traite le trafic, le second reste en attente et prend le relais automatiquement en cas de panne.

**Rôle de la VIP** : l'adresse IP virtuelle est utilisée par le client pour accéder au service ; elle bascule automatiquement d'un nœud HAProxy à l'autre afin d'assurer la continuité du service.

**Configuration des fichiers** : Heartbeat repose sur plusieurs fichiers (ha.cf, authkeys, haresources) permettant de définir les nœuds du cluster, l'authentification et les ressources à basculer (VIP et service HAProxy).

/etc/ha.d/ha.cf

```
GNU nano 8.4
keepalive 2
deadtime 10
warntime 5
initdead 15
udpport 694
ucast ens33 192.168.0.12
node HAProxy
node HAProxy2
auto_failback on
```

/etc/ha.d/authkeys

```
GNU nano 8.4
auth 1
1 sha1 elmonstro
```

/etc/ha.d/haresources

```
GNU nano 8.4
HAProxy IPaddr::192.168.0.10/24/ens33 HAProxy
```

ip a → VIP visible sur HAProxy1

```
root@HAProxy:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:9e:25:4e brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c299e254e
    inet 192.168.0.129/24 brd 192.168.0.255 scope global dynamic noprefixroute ens33
        valid_lft 1184sec preferred_lft 959sec
    inet 192.168.0.11/24 brd 192.168.0.255 scope global secondary ens33
        valid_lft forever preferred_lft forever
    inet 192.168.0.10/24 brd 192.168.0.255 scope global secondary ens33:0
        valid_lft forever preferred_lft forever
    inet6 fe80::183a:6a80:1e4a:4302/64 scope link
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:9e:25:58 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    altname enx000c299e2558
    inet 172.20.0.11/16 brd 172.20.255.255 scope global ens37
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe9e:2558/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

systemctl status haproxy

```
root@HAProxy:~# systemctl status haproxy
• haproxy.service - HAProxy Load Balancer
   Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-21 02:04:41 CEST; 10min ago
   Invocation: a264ca5f12654086a19faf4e9f9ad482
     Docs: man:haproxy(1)
           file:/usr/share/doc/haproxy/configuration.txt.gz
   Main PID: 1248 (haproxy)
   Status: "Ready."
     Tasks: 3 (limit: 2255)
  Memory: 78.8M (peak: 79.8M)
     CPU: 563ms
   CGroup: /system.slice/haproxy.service
           └─1248 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock
             └─1278 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock

avril 21 02:04:40 HAProxy systemd[1]: Starting haproxy.service - HAProxy Load Balancer...
avril 21 02:04:41 HAProxy haproxy[1248]: [NOTICE] (1248) : New worker (1278) forked
avril 21 02:04:41 HAProxy systemd[1]: Started haproxy.service - HAProxy Load Balancer.
avril 21 02:04:41 HAProxy haproxy[1248]: [NOTICE] (1248) : Loading success.
root@HAProxy:~#
```

## Étape 5 – Tests de bascule

### Test 1 – Load Balancing OK

Plusieurs curl http://192.168.0.10

Alternance SERV-WEB1 / SERV-WEB2

```
C:\Users\pc47>curl http://192.168.0.10
srvweb1

C:\Users\pc47>curl http://192.168.0.10
srvweb2
```

### Test 2 – Panne HAProxy1

VM HAProxy1 arrêtée

```
valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:9e:25:4e brd ff:ff:ff:ff:ff:ff
altname enp2s1
altname enx000c299e254e
inet 192.168.0.129/24 brd 192.168.0.255 scope global dynamic noprefixroute ens33
    valid_lft 1792sec preferred_lft 1567sec
inet 192.168.0.11/24 brd 192.168.0.255 scope global secondary ens33
    valid_lft forever preferred_lft forever
inet 192.168.0.10/24 brd 192.168.0.255 scope global secondary ens33:0
    valid_lft forever preferred_lft forever
inet6 fe80::103a:6a80:1e4a:4302/64 scope link
    valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:9e:25:58 brd ff:ff:ff:ff:ff:ff
altname enp2s5
altname enx000c299e2558
inet 172.20.0.11/16 brd 172.20.255.255 scope global ens37
    valid_lft forever preferred_lft forever
inet6 fe80::20c:29ff:fe9e:2558/64 scope link proto kernel_ll
    valid_lft forever preferred_lft forever
root@HAProxy:~# systemctl stop heartbeat
root@HAProxy:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:9e:25:4e brd ff:ff:ff:ff:ff:ff
altname enp2s1
altname enx000c299e254e
inet 192.168.0.129/24 brd 192.168.0.255 scope global dynamic noprefixroute ens33
    valid_lft 1759sec preferred_lft 1534sec
inet 192.168.0.11/24 brd 192.168.0.255 scope global secondary ens33
    valid_lft forever preferred_lft forever
inet6 fe80::103a:6a80:1e4a:4302/64 scope link
    valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:9e:25:58 brd ff:ff:ff:ff:ff:ff
altname enp2s5
altname enx000c299e2558
inet 172.20.0.11/16 brd 172.20.255.255 scope global ens37
    valid_lft forever preferred_lft forever
inet6 fe80::20c:29ff:fe9e:2558/64 scope link proto kernel_ll
    valid_lft forever preferred_lft forever
```

## ip a → VIP sur HAProxy2

```
root@HAProxy2:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:74:79:dc brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c297479dc
    inet 192.168.0.128/24 brd 192.168.0.255 scope global dynamic noprefixroute ens33
        valid_lft 1108sec preferred_lft 883sec
    inet 192.168.0.12/24 brd 192.168.0.255 scope global secondary ens33
        valid_lft forever preferred_lft forever
    inet 192.168.0.10/24 brd 192.168.0.255 scope global secondary ens33:0
        valid_lft forever preferred_lft forever
    inet6 fe80::2c19:ba81:358a:fa02/64 scope link
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:74:79:e6 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    altname enx000c297479e6
    inet 172.20.0.12/16 brd 172.20.255.255 scope global ens37
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe74:79e6/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

curl client toujours OK

```
C:\Users\pc47>
C:\Users\pc47>curl http://192.168.0.10
srvweb1

C:\Users\pc47>curl http://192.168.0.10
srvweb2

C:\Users\pc47>
```

## Test 4 – Retour à la normale

### Redémarrage HAProxy1

### VIP revient sur HAProxy1

### Load balancing OK

```
root@HAProxy:~# systemctl status haproxy
● haproxy.service - HAProxy Load Balancer
   Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-21 02:33:41 CEST; 16s ago
     Invocation: 6bcf46fb8a5b4fcd8d871093dfb19d50
       Docs: man:haproxy(1)
            file:/usr/share/doc/haproxy/configuration.txt.gz
   Main PID: 1211 (haproxy)
      Status: "Ready."
        Tasks: 3 (limit: 2255)
     Memory: 75.9M (peak: 76.1M)
        CPU: 174ms
   CGroup: /system.slice/haproxy.service
           └─1211 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock
             1234 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/haproxy.pid -S /run/haproxy-master.sock

avril 21 02:33:40 HAProxy systemd[1]: Starting haproxy.service - HAProxy Load Balancer...
avril 21 02:33:41 HAProxy haproxy[1211]: [NOTICE] (1211) : New worker (1234) forked
avril 21 02:33:41 HAProxy systemd[1]: Started haproxy.service - HAProxy Load Balancer.
avril 21 02:33:41 HAProxy haproxy[1211]: [NOTICE] (1211) : Loading success.
root@HAProxy:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:9e:25:4e brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c299e254e
    inet 192.168.0.129/24 brd 192.168.0.255 scope global dynamic noprefixroute ens33
        valid_lft 1779sec preferred_lft 1554sec
    inet 192.168.0.11/24 brd 192.168.0.255 scope global secondary ens33
        valid_lft forever preferred_lft forever
    inet 192.168.0.10/24 brd 192.168.0.255 scope global secondary ens33:0
        valid_lft forever preferred_lft forever
    inet6 fe80::183a:6a80:1e4a:4302/64 scope link
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:9e:25:58 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    altname enx000c299e2558
    inet 172.20.0.11/16 brd 172.20.255.255 scope global ens37
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe9e:2558/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
root@HAProxy:~#
```

# Conclusion

Ce projet a permis de mettre en œuvre une infrastructure web en haute disponibilité reposant sur l'association de HAProxy et Heartbeat. HAProxy assure la répartition de charge entre plusieurs serveurs web grâce à une méthode round-robin, tandis que Heartbeat garantit la continuité du service via une architecture actif/passif et une adresse IP virtuelle (VIP).

Les différents tests réalisés ont confirmé le bon fonctionnement du système : la répartition de charge s'effectue correctement entre les serveurs backend, et la VIP bascule automatiquement vers le nœud secondaire en cas de panne du nœud principal. Cette solution permet ainsi de maintenir l'accès au service sans interruption du point de vue client.

Ce projet a permis d'approfondir les notions de haute disponibilité, de tolérance aux pannes et de supervision de services Linux. Il a également mis en évidence l'importance de la rigueur dans la configuration des fichiers et de la cohérence du réseau. Des améliorations pourraient être apportées, comme la sécurisation des services ou l'ajout de mécanismes de supervision avancée, afin de se rapprocher davantage d'une infrastructure professionnelle.



